

### 3. Phase 3: Unsupervised Phenotype Stratification (N = 699)

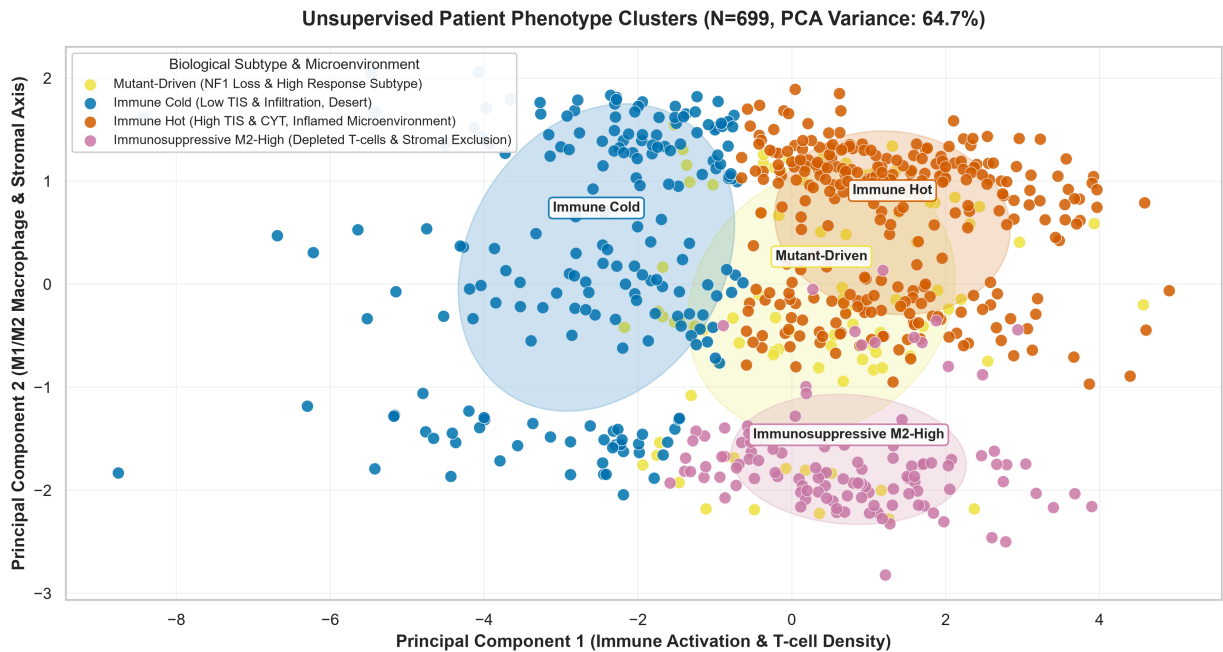
 **Analytical Methodology & Rationale**

- What is being done:** Applying K-Means clustering ( $K = 4$ ) to feature matrices and generating 2D Principal Component projections.
- Why we are doing it:** Unsupervised clustering discovers natural biological patient subgroups without outcome bias. Embedding visual projections is preferred over raw tables for slide presentation.
- What question it answers:** What distinct patient clusters emerge from multi-dimensional biological profiling?

#### Unsupervised Phenotype Cluster Summary

| Cluster ID | Biological Phenotype Subtype                                     | Patient Count (N) | Cohort Share | Response Rate |
|------------|------------------------------------------------------------------|-------------------|--------------|---------------|
| Cluster 0  | Mutant-Driven (NF1 Loss & High Response Subtype)                 | 89                | 12.7%        | <b>61.5%</b>  |
| Cluster 1  | Immune Cold (Low TIS & Infiltration, Desert)                     | 198               | 28.3%        | <b>35.4%</b>  |
| Cluster 2  | Immune Hot (High TIS & CYT, Inflamed Microenvironment)           | 289               | 41.3%        | <b>38.9%</b>  |
| Cluster 3  | Immunosuppressive M2-High (Depleted T-cells & Stromal Exclusion) | 123               | 17.6%        | <b>45.2%</b>  |

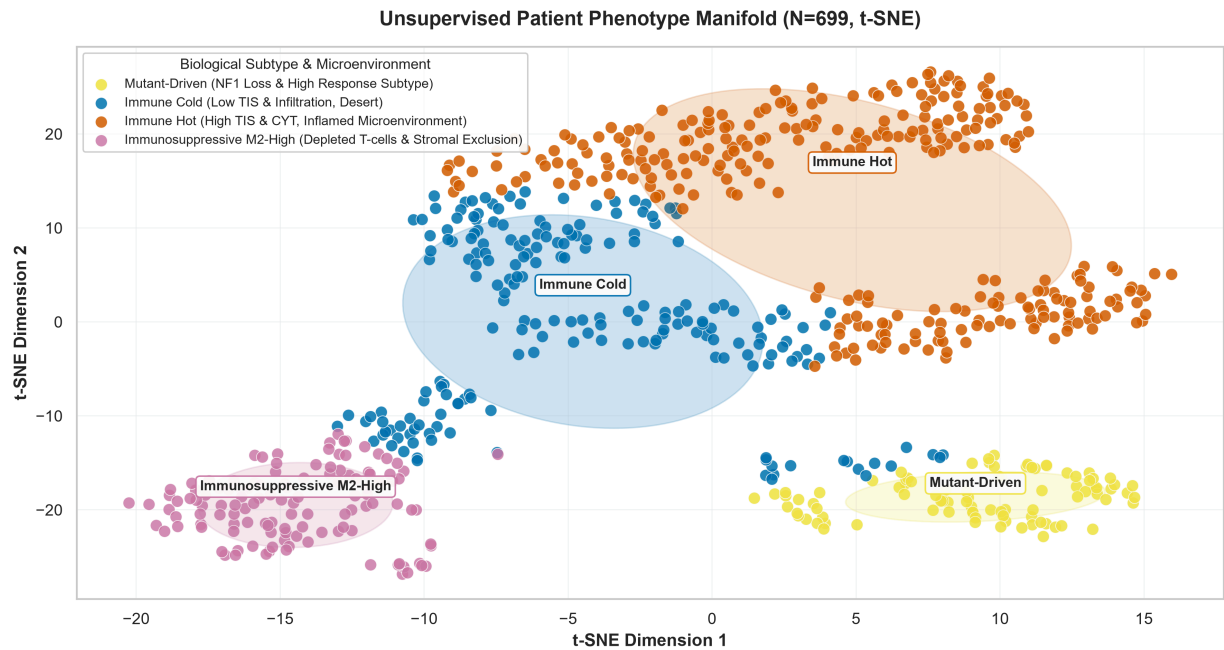
#### Unsupervised Phenotype Cluster Projection (2D PCA)



 **Figure Interpretation: 2D Principal Component Cluster Projection**

- What this plot shows:** 2D Principal Component Projection of  $N = 699$  patients colour-coded by their multi-modal K-Means phenotype cluster ( $K = 4$ ). Shaded confidence ellipses mark cluster boundaries.
- Axis 1 (Horizontal):** Principal Component 1 captures immune activation and lymphocytic T-cell density (separating Inflamed Hot vs Desert Cold tumours).
- Axis 2 (Vertical):** Principal Component 2 captures macrophage polarisation (M1/M2 ratio) and stromal CAF exclusion.
- Clinical Value:** Discovers discrete patient subgroups with distinct treatment response profiles without relying on biased outcome labels.

#### Unsupervised Phenotype Manifold (t-SNE Projection)



#### Figure Interpretation: Non-Linear t-SNE Cluster Manifold

- **What this plot shows:** 2D t-SNE non-linear manifold projection of the 9-feature patient space ( $N = 699$ ), colour-coded by the K-Means cluster labels assigned in full 9-dimensional feature space.
- **Non-Linear Topology:** t-SNE (perplexity=50) preserves local patient neighbourhood structure and non-linear biomarker interactions across the 9 multi-modal clustering features (TIS, CYT, CD8 T-cells, M1/M2 Macrophages, CAFs, BRAF / NRAS / NF1 mutations). Natural within-cluster scatter reflects genuine continuous variation within each immune phenotype.

### Key Takeaways & Student Summary

- **Distinct Patient Groups:** K-Means clustering partitioned  $N = 699$  patients (full cohort) into four biological subgroups; within the ICI-treated sub-cohort ( $N = 326$ ), response rates range from **35.4% to 61.5%**.
- **Highest Response Group:** The **Mutant-Driven (NF1 Loss & High Response Subtype)** subgroup achieves the highest response rate (61.5%), benefiting from favorable immune activation and high driver mutation burden.
- **Treatment-Resistant Subgroup:** The **Immune Cold (Low TIS & Infiltration, Desert)** subgroup exhibits the lowest response rate (35.4%), highlighting the need for targeted combination therapies beyond single-agent PD-1 blockade.

#### Student-Friendly Phase 3 Summary

Phase 3 performed unsupervised multi-dimensional clustering to discover natural biological patient subgroups without relying on outcome labels:

1. **Four Distinct Phenotypes:** K-Means clustering ( $K = 4$ ) partitioned patients into *Immune Hot (High TIS & CYT, Inflamed Microenvironment)*, *Immune Cold (Low TIS & Infiltration, Desert)*, *Immunosuppressive M2-High (Depleted T-cells & Stromal Exclusion)*, and *Mutant-Driven (NF1 Loss & High Response Subtype)* phenotypes across 9 biomarker axes.
2. **Wide Response Rate Divergence:** Clinical response rates varied markedly across clusters, demonstrating that unselected cohort averages mask distinct biological subgroups.
3. **Dimensionality Projections:** 2D PCA and non-linear t-SNE projections confirm clear spatial separation, with PC1 capturing T-cell inflammation and PC2 capturing myeloid/stromal exclusion.
4. **Clinical Takeaway:** Identifying a patient's biological phenotype provides the foundation for targeted routing rather than applying a single uniform treatment protocol.